

CIS 121 Introduction to Programming

Problem Set – Nested If and compound relational conditions. For each problem, create an IPO chart and c++ code file.

1. The input to the problem is the quantity of widgets and customer status. You set the price based upon quantity and status using the table below. Your program should determine the price to charge based on the schedule below. Calculate the extended price. Calculate tax at 7%. Display the extended price, tax amount, and total.

Quantity	Status	Price
>10000	A	\$10
>10000	B	\$12
5000 to 10000	C	\$20
5000 to 10000	D	\$22
Below 5000	Any	\$30

Input	Process	Output
cStatus	<pre> switch (cStatus) {case 'A': if (widgetQty > 10000) price = 10.00; else price = 30.00; break; case 'B': if (widgetQty > 10000) price = 12.00; else price = 30.00; break; case 'C': if (widgetQty >= 5000 && widgetQty <= 10000) price = 20.00; else price = 30.00; break; case 'D': if (widgetQty >= 5000 && widgetQty <= 10000) price = 22.00; else price = 30.00; break; default: price = 30.00; // any other status break; </pre>	
widgetQty		
Tax		Extended Price:
Extprice		Tax Amount:
total		Total:
price		

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2. Enter a part number of the following (10, 99, 55, 70, 50). Also enter the quantity. Determine the cost per unit using the table below. Then calculate the total cost. Display the part number, cost per unit and total.

Part	Quantity	Cost Per Unit
10	> 1000	1.00
99	> 500	2.00
All others	All others	5.00

Input	Process	Output
	<pre> if (partNumber == 10 && quantity > 1000) { costPerUnit = 1.00; } else if (partNumber == 99 && quantity > 500) {costPerUnit = 2.00; } else { costPerUnit = 5.00; // all other parts and quantities } </pre>	
PNumber	Total = costPerUnit * qty	Part Number:
qty		Cost Per Unit:
Total		Total:
costPerUnit		

3. Allow the user to enter number of concert tickets and location code (H, L). The price per ticket depends on the volume and location (see below). Display the number of tickets, price per ticket and the total cost.

Volume is greater than 25 or location is H cost per ticket is \$30.00

Volume is greater than 10 (10 to 24) or location is L cost per ticket is \$40.00

All other quantities or locations are \$50.00

Input	Process	Output
numberOfTickets	if (tickets > 25 locationCode == 'H') { pricePerTicket = 30.00; } else if (tickets > 10 locationCode == 'L') { pricePerTicket = 40.00; } else { pricePerTicket = 50.00; }	n
locationCode	Total= pricePerTicket* numberOfTickets	Number of Tickets:
pricePerTicket		Price Per Ticket:
Total		Total Cost:

4. Allow the user to enter equipment code of a rental and a code indicating half day or full day. Determine the cost of the rental. Display the rental cost

Equipment Code	Day	Cost
A	F	10.00
A	H	15.00
B	F	20.00
B	H	35.00
C	H	40.00
C	F	45.00
All others	All others	50.00

Input	Process	Output
eCode	if (equipCode == 'A' && dayCode == 'F') { cost = 10.00; } else if (equipCode == 'A' && dayCode == 'H') { cost = 15.00;} else if (equipCode == 'B' && dayCode == 'F') { cost = 20.00; }	

	<pre> else if (equipCode == 'B' && dayCode == 'H') { cost = 35.00; } else if (equipCode == 'C' && dayCode == 'H') { cost = 40.00; } else if (equipCode == 'C' && dayCode == 'F') {cost = 45.00; } else { cost = 50.00; // all other combinations } </pre>	
dCode		Renter Last Name:
Cost		Rental Cost:

5. You need to display the gross salary for an employee. They input a job code and Hours. First, determine the rate of pay based on job code and hours (see table below). Next, compute gross pay (hrs * rate). No overtime pays.

Job Code	Hours	Rate of Pay
L	> 40	50.00
L	<=40	40.00
J	>60	100.00
J	<=60	75.00
A	>40	25.00
A	<=40	20.00

Input	Process	Output
jobCode	<pre> if (jobCode == 'L') { rate = (hours > 40) ? 50.00 : 40.00; } else if (jobCode == 'J') { rate = (hours > 60) ? 100.00 : 75.00; } </pre>	

	<pre> else if (jobCode == 'A') { rate = (hours > 40) ? 25.00 : 20.00; } else { cout << "Invalid job code." << endl; return 1; } </pre>	
hrs	Hrs*rate= gross salary	Employee Last Name:
GPay		Job Code:
Rate		Rate Per Hour:
Lname		Gross Salary: